

Polymorph, a minimal input procedural modeling tool for rapid prototyping of stop motion puppets at LAIKA

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ABSTRACT

Polymorph is a minimal input voxel-poly toolset for interactive creation of stop motion puppet head and neck assemblies. At its core it generates procedural 3D print-ready geometry for replacement facial animation.

CCS CONCEPTS

• **Software and its engineering** → **Software design techniques**; **Requirements analysis**.

KEYWORDS

rapid prototyping, pipeline, procedural modeling

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1 INTRODUCTION

A stop motion puppet head assembly is a complex region of a puppet, comprised of many parts. An assembly includes eyes that pitch, yaw and translate, smoothly rotating lids, removable parts for maintenance, tension access to internal joints, attachable parts for hand dressed elements eg. hair, multiple tie-down points to control the puppet with external rigging, labeling for part version tracking (texture-based and physically embossed), registrational elements and magnets for facial replacement, sprue connections for automated print processing, different mechanical allowances for various print materials, hardware attachment for hats or glasses, and various other parts. Altogether, the design of a puppet head assembly can require upwards of 70 parts inside a single head, the smallest of which is 3/4 of an inch in width.

Because different physical facial expressions will be applied to the head assembly 24 times per second and displayed on a 65-foot-wide screen, the parts must come together with high precision. A discrepancy in facial registration measuring one third of the thickness of a human hair is enough to result in visible jittering in the final product. Polymorph is a toolset that takes a minimum

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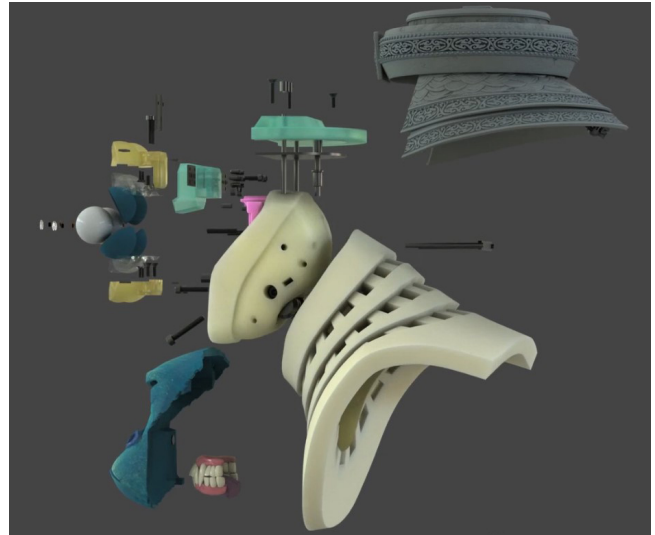


Figure 1: Example of Puppet Head Assembly Parts

of 6 input meshes, and procedurally and parametrically creates an entire puppet head assembly. The user can modify parameters or entire hardware configurations in real-time for greater design flexibility, turning months of work into days. Polymorph ensures precision head assembly results and even generates supporting printable parts such as keys for molding silicone necks, lid cutting parts, vacuum forming lid-sleeve blocks, etc.

2 CHALLENGES

Every LAIKA puppet is unique, featuring a wide range of proportional and scale differences that impart unique spatial constraints to the internal hardware needed. Puppets span a gamut of configurations, from having giant necks which require large joints and carefully planned silicone thicknesses and internal support structures, to pencil-necked characters requiring tiny custom joints or metal parts. This variability makes traditional puppet builds take months to develop to a final state. In the prior pipeline, changing a single part would trigger a cascade of revisions across several neighboring parts. Without dynamic relationships, the changes all needed manual management. This was not only quite difficult to manage, but error-prone even for a seasoned modeler. Such errors are costly because of the turnaround times and labor involved in 3D printing, processing, or molding and casting in silicone. While parametric modeling tools often found in CAD packages have some of the needed features, they haven't been a great overall fit for

LAIKA's puppet head needs. LAIKA uses a traditional animation pipeline for facial animation, which means CG modelers need to be frequently working with data from Maya on the rigging and animation side. There is also a common need to make expedient custom edits to any given part, which can be cumbersome in a traditional parametric system. Finally, it has been difficult to find talented digital sculptors for our CG modeling positions who also have experience in CAD based modeling applications.

3 THE SOLUTION

That is where Polymorph comes into play. This Houdini-based toolset leverages the benefits of parametric part design, custom tailored to LAIKA's process and needs, while maintaining a polygon-based workflow. Polymorph's first operation converts portions of the head and neck into voxels to facilitate hollow shelling of surfaces needed to generate initial parts. Users can easily control registrational elements, screws and magnets via interactive handles in viewport. Other components dynamically constrain to parts via curves generated from surface contours. Polymorph includes a system to smooth boolean intersections for parts behind the eyeballs or under the neck joint for creating a smooth range of motion. Users can make all of their design decisions interactively in Polymorph, and with a single click, bring their work seamlessly into Maya. A Python script in Maya imports Houdini's geometry and adds previs animation to the parts, so that users can quickly assess their builds. Initially it was thought this would all be done in Houdini, but employees reported that Maya's viewport and camera controls were superior for navigating spaces between parts, so the tool was adapted to transfer data quickly between the two software packages. Given the varied and unpredictable performance needs of a puppet, inevitably there are instances where we need the ability to customize the shapes of certain parts. Editing the final result of a procedural process can be destructive, as future parametric changes made upstream would destroy any polygon editing done by hand. To remain fully non-destructive without branching away from proceduralism and still allowing hand editable geometry, two solutions were implemented. Firstly, Polymorph gives its users the freedom to insert custom meshes "midstream" into the algorithm at carefully chosen points, so parts can be hand-crafted and part of a dynamic system simultaneously. Secondly, a boolean tool was developed in Maya to allow any final boolean steps to take place on build parts non-destructively. The boolean system itself has name-based contextual features to mirror, smooth, or create accurate offsets of geometry at run-time, as well as troubleshoot boolean failure steps.

3.1 Part Fitting

When presented with variable unknown shapes that must fit together, one must deal with incompatibilities. To address this, spatial solutions were designed to support universal part fitting. One example is for parts with undercuts, a common feature preventing two parts from physically assembling. For these problematic features, the outer silhouette of a part is converted into a simplified curve which then projects backwards at the chosen angle of removal of the attachment hardware, and is converted into a band of geometry to crop its mating part. This affords the user flexibility in selecting

an ideal attachment angle, knowing that any choice will result in successful part fitting.

4 ADAPTABILITY

At any point, if LAIKA decides to change 3D, find ways to make hardware assemblies more efficient, or change slicing software (causing the mechanical allowances of certain materials to be affected), Polymorph allows for such changes to easily propagate into all puppet builds. In the past when such changes have been necessary, all puppets needed to be re-visited one by one and hand edited to unify similar updates. This new build approach, along with the ability to quickly compare any step of any part across multiple builds, has brought some unexpected benefits. It's become much easier to find common spatial constraints across various builds. This has led to optimizations in Polymorph which end up improving all builds, instead of just one. Another unplanned benefit involved data footprint, with procedural modeling systems allowing for much smaller file sizes.

5 CONS

Some features that are included in a CAD based package, such as simply placing a screw are not available in Houdini. This necessitated the creation of a modular screw rig capable of supporting a plethora of screws. Screws can change head type, thread count, or length. When snapped to a surface, the head will automatically countersink itself if flat or align to the surface if hexagonal. Users can choose between a threaded boolean to cut through parts, or opt for a simple tap or thru-hole boolean. For magnets, a similar system was devised with options for magnet housing shapes to allow more flexible placement than surface-snapping, and customizable drain holes to clear 3D print support material.

6 CURRENT STATE

Polymorph is still in its early stages, but off to a promising start. It has been used to create eight digital builds of various proportions, two of which have been 3D printed and assembled for production of LAIKA's current film. The part fitting and build quality showed a noted improvement from our prior hand modeled initial build prints. In the future it would be ideal to take a pass at performance optimization and continue to add more solutions to common spatial constraints. Currently, performance can be controlled by a global voxel size which balances between accuracy and real-time interaction.

7 CONCLUSION

LAIKA has always embraced combining the old with the new. This has manifested as digitization alongside careful handcraft eg. 6,000 year old bronze metal casting techniques, used with state of the art 3D printing processes. In some way, we all strive to strike a balance between tradition and innovation. Polymorph attempts to stay true to those aspirations, by offering the best of a parametric workflow, while retaining the ability to manually adjust polygons for immediate results.